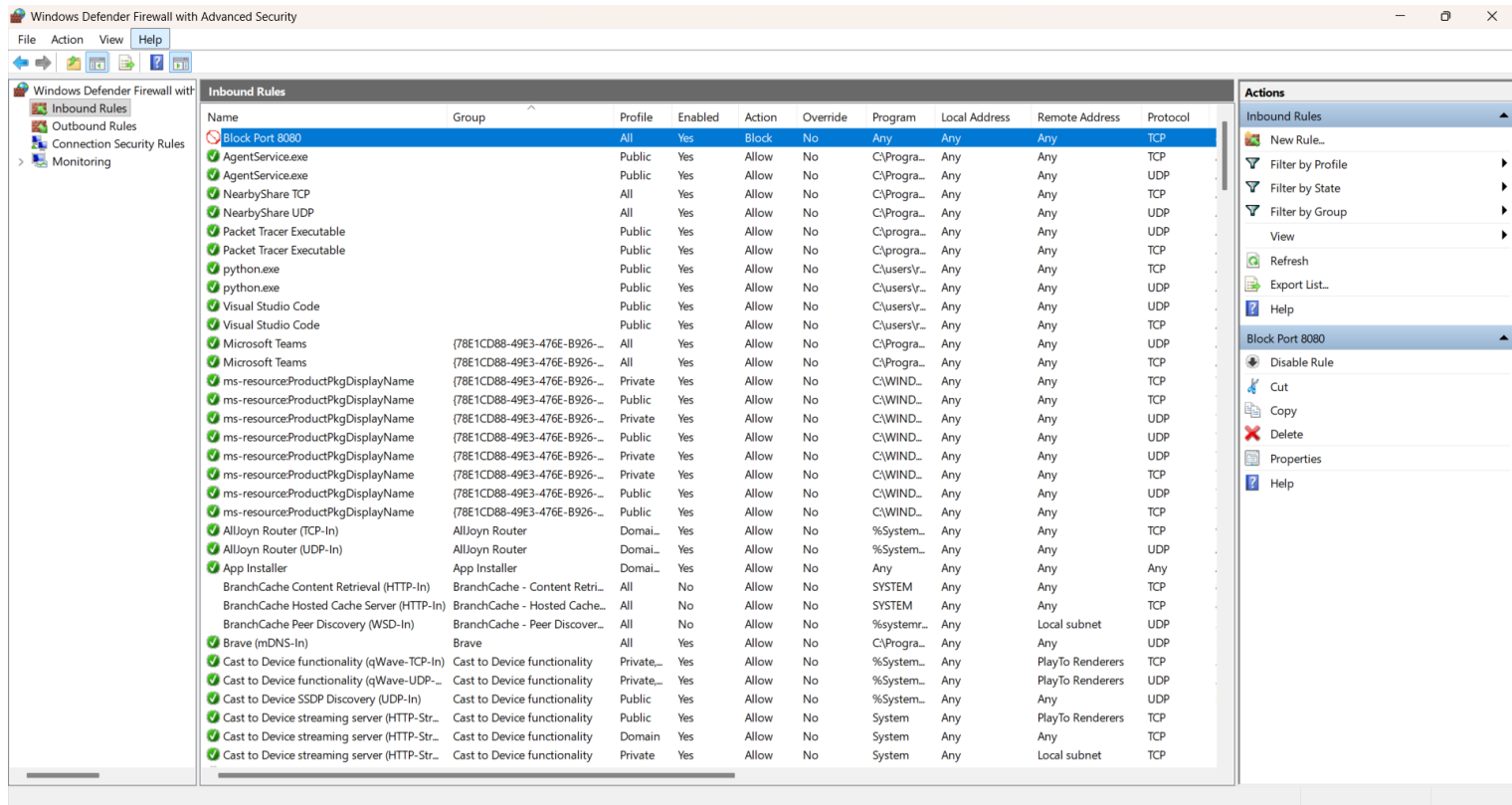
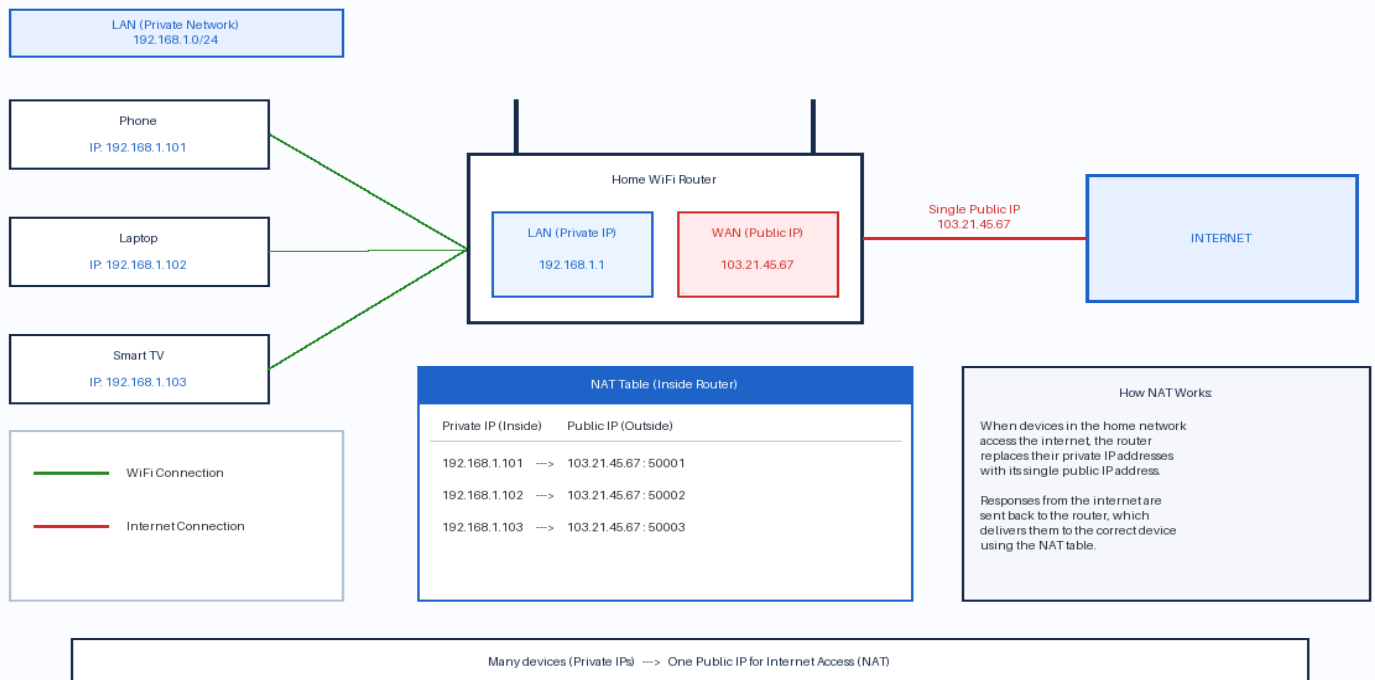


TASK-1



TASK-2

NAT (Network Address Translation) in Home WiFi Router



TASK-3

Since I can't access your router's admin panel, here's an assignment-ready example showing how to configure **port filtering** using a typical Wi-Fi router or router simulator.

Configure Port Filtering to Block Outgoing SMTP (Port 25)

Steps

1. Open a web browser and log in to your router's admin page (e.g., <http://192.168.1.1> or <http://192.168.10.1>).
2. Enter the administrator username and password.
3. Navigate to **Security, Firewall, Access Control**, or **Port Filtering** (the name varies by router model).
4. Click **Add New Rule** or **Create Rule**.
5. Configure the rule as follows:

Setting	Value
Rule Name	Block SMTP
Direction	Outbound
Protocol	TCP
Destination Port	25
Action	Block / Deny
Status	Enabled

6. Click **Save** or **Apply**.
7. Restart the router if prompted.

Example Configuration

Rule Name	Direction	Protocol	Port	Action	Status
Block SMTP	Outbound	TCP	25	Block	Enabled

Explanation (2 Sentences)

Blocking **outgoing traffic on port 25 (SMTP)** helps prevent infected computers or malware on the network from sending spam emails directly to external mail servers. This improves network security and reduces the risk of the network being used to distribute spam or malicious emails.

TASK-4

PACKET FILTERING RULE SET						
Rule No.	Source IP	Destination IP	Protocol	Destination Port	Action	Description
1	192.168.1.0/24	Any	TCP	80	ALLOW	Allow HTTP from LAN
2	192.168.1.0/24	Any	UDP	53	ALLOW	Allow DNS from LAN
3	10.0.0.0/24	Any	TCP	22	ALLOW	Allow SSH from 10.0.0.0/24
4	Any	Any	TCP	25	DENY	Block SMTP (Email)
5	Any	Any	Any	Any	DENY	Deny all other traffic

SAMPLE PACKETS TO EVALUATE

Packet No.	Description	Source IP	Destination IP	Protocol	Destination Port
1	HTTP request	192.168.1.5	172.217.166.78	TCP	80
2	SSH request	10.0.0.10	34.201.23.1	TCP	22
3	DNS query	192.168.1.7	8.8.8.8	UDP	53

EVALUATION

Packet No.	Result	Matched Rule	Explanation
1	ALLOW	Rule 1	Source 192.168.1.5 is in 192.168.1.0/24 and destination port is 80 (HTTP) → Allowed
2	ALLOW	Rule 3	Source 10.0.0.10 is in 10.0.0.0/24 and destination port is 22 (SSH) → Allowed
3	ALLOW	Rule 2	Source 192.168.1.7 is in 192.168.1.0/24 and destination port is 53 (DNS) → Allowed

TASK-5

How Instagram or Zomato Use Firewalls and Packet Filtering

1. Blocking Unauthorized Access

- Firewalls allow only approved ports and services (such as HTTPS on port 443) while blocking unauthorized connection attempts. This helps prevent hackers from accessing servers through unused or unsafe ports.

2. Filtering Malicious Traffic

- Packet filtering checks incoming network packets based on IP addresses, ports, and protocols. Suspicious traffic, such as packets from known malicious IP addresses or abnormal requests, can be blocked before reaching the application servers.

3. Protection Against DDoS and Abuse

- Firewalls can detect and limit excessive requests from the same IP address, helping reduce the impact of Distributed Denial-of-Service (DDoS) attacks, spam, and automated bots. This keeps services like Instagram and Zomato available for legitimate users.

Summary

Security Feature	Purpose	Example
Firewall Rules	Block unauthorized access	Allow HTTPS (443), block unused ports
Packet Filtering	Inspect and filter network traffic	Block suspicious IP addresses or malicious packets
Traffic Monitoring & Rate Limiting	Prevent DDoS attacks and misuse	Limit excessive login or API requests from a single IP